

Title:

TSB

SENSOR INSPECTION FOR AIR CONDITIONING SYSTEM

Models:

'89 – Current All Models

Introduction This service bulletin contains inspection procedures to more precisely confirm proper operation of the following temperature sensors of the air conditioning system. Follow the procedures in this service bulletin when inspecting these sensors. These contents will be reflected in future repair manuals.

- Room Temperature Sensor
- Ambient Temperature Sensor
- Air Duct Sensor
- Evaporator Temperature Sensor
- Solar Sensor
- Room Humidity Sensor

Applicable Vehicles • All 1989 – Current model year **Toyota** vehicles.

Warranty Information

OP CODE	DESCRIPTION	TIME	OFF	T1	T2
N/A	Not Applicable to Warranty	–	–	–	–



Inspection
Procedure

1. Inspect Room Temperature Sensor.

A. Measure the sensor resistance.

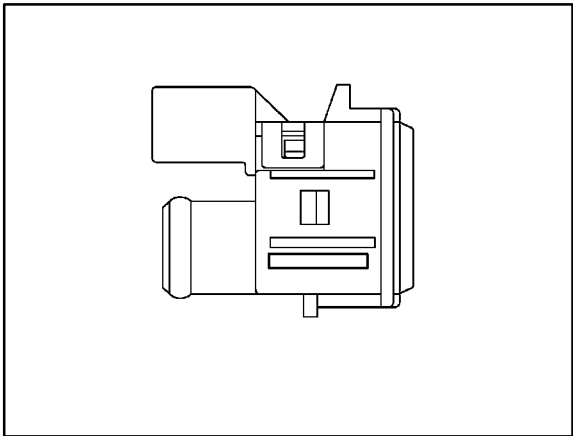
Resistance Value at 77°F (25°C)	1700 +/- 85Ω
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NOTE:

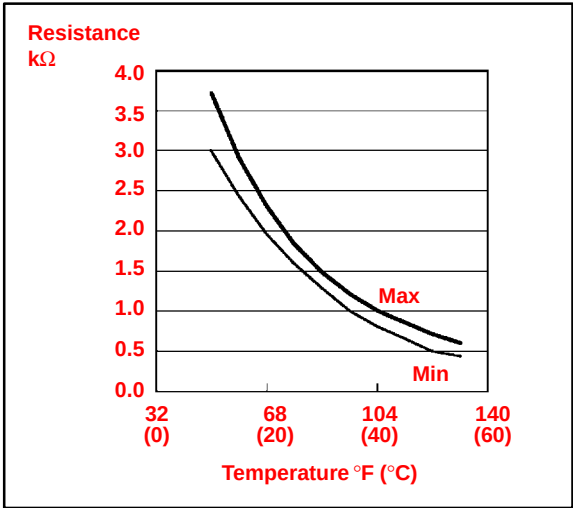
- Even slightly touching the sensor may change the resistance value. Be sure to hold the connector of the sensor.
- When measuring, the sensor temperature must be the same as the ambient temperature.

HINT:

As the temperature increases, the resistance decreases.



TEMPERATURE °F (°C)	SPECIFICATION kΩ
50 (10)	3.00 to 3.73
59 (15)	2.45 to 2.88
68 (20)	1.95 to 2.30
77 (25)	1.60 to 1.80
86 (30)	1.28 to 1.47
95 (35)	1.00 to 1.22
104 (40)	0.80 to 1.00
113 (45)	0.65 to 0.85
122 (50)	0.50 to 0.70
131 (55)	0.44 to 0.60
140 (60)	0.36 to 0.50



**Inspection
Procedure
(Continued)**

**2. Inspect Ambient
Temperature Sensor.**

- A. Measure the sensor resistance according to the selected graph (specification).

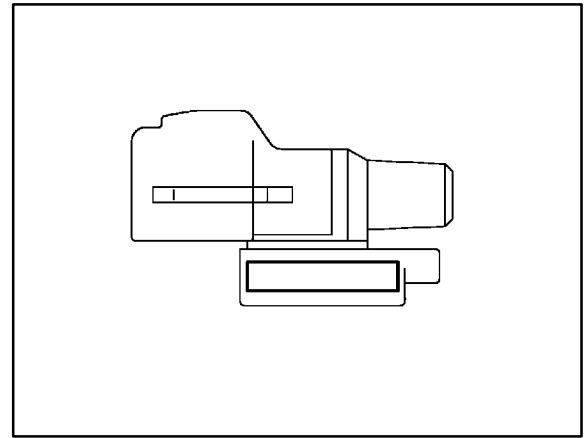
Resistance Value at 77°F (25°C)	1700 +/- 85Ω
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NOTE:

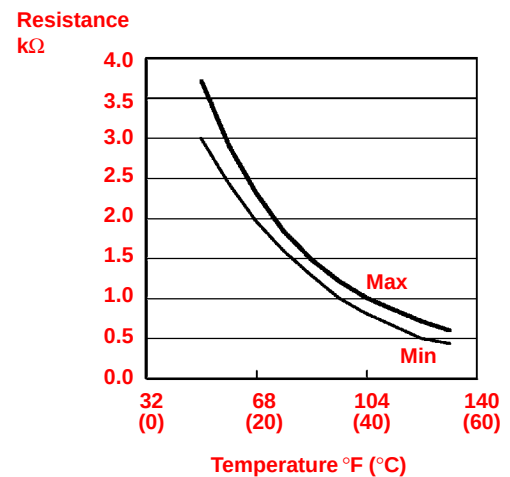
- Even slightly touching the sensor may change the resistance value. Be sure to hold the connector of the sensor.
- When measuring, the sensor temperature must be the same as the ambient temperature.

HINT:

As the temperature increases, the resistance decreases.



TEMPERATURE °F (°C)	SPECIFICATION kΩ
50 (10)	3.00 to 3.73
59 (15)	2.45 to 2.88
68 (20)	1.95 to 2.30
77 (25)	1.60 to 1.80
86 (30)	1.28 to 1.47
95 (35)	1.00 to 1.22
104 (40)	0.80 to 1.00
113 (45)	0.65 to 0.85
122 (50)	0.50 to 0.70
131 (55)	0.44 to 0.60
140 (60)	0.36 to 0.50



**Inspection
Procedure**
(Continued)

3. Inspect Air Duct Sensor.

- A. Measure the sensor resistance according to the table and graph (specification).

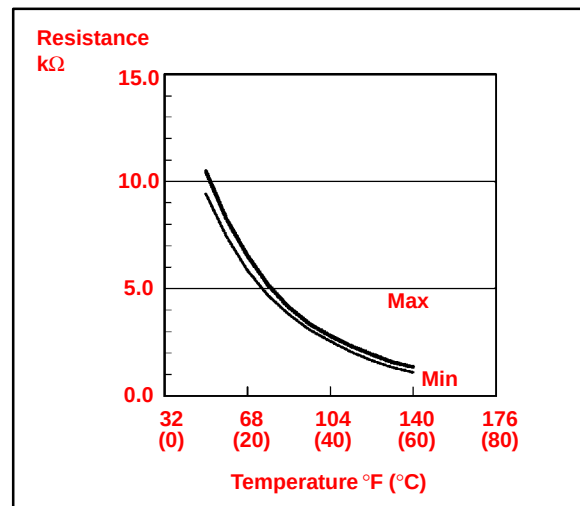
NOTE:

- Even slightly touching the sensor may change the resistance value. Be sure to hold the connector of the sensor.
- When measuring, the sensor temperature must be the same as the ambient temperature.

HINT:

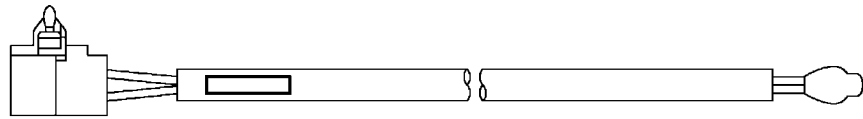
As the temperature increases, the resistance decreases.

TEMPERATURE °F (°C)	SPECIFICATION kΩ
50 (10)	9.48 to 10.49
59 (15)	7.50 to 8.28
68 (20)	5.95 to 6.57
77 (25)	4.77 to 5.25
86 (30)	3.85 to 4.21
95 (35)	3.12 to 3.40
104 (40)	2.53 to 2.79
113 (45)	2.06 to 2.30
122 (50)	1.69 to 1.91
131 (55)	1.39 to 1.59
140 (60)	1.15 to 1.33



Inspection
Procedure
(Continued)

4. Inspect Evaporator Temperature Sensor.



Select the appropriate graph (specification) using the following table.

NOTE:

Please inspect the sensors for model years not indicated by this bulletin, according to the instructions in the applicable repair manual.

MODEL	MODEL YEAR	COMMENTS	PART NUMBER	GRAPH
4Runner	1989 – 1993		88625–89103	2
	1994 – 1995		88625–35020	2
	1996 – 2002		88625–3A090	2
	2003		88625–35050	3
	2003 – 2005		88625–35090	4
Avalon	1995 – 1999		88625–33070	2
	2000 – 2005		88625–47011	2
Camry	1992 – 1996	CBU	88625–33070	2
		NAP		
	1997 – 2001	CBU	88625–33140	2
		NAP		
	2001	NAP	88625–17130	2
		CBU		
	2002 – 2005	All	88625–33170	3
Celica	1994 – 1995	5SFE	88625–20270	2
	1994 – 1999	7AFE or GT	88625–60050	2
	1998 – 1999	ST		
	2000 – 2005	All	88625–20510	2
Corolla	1993 – 2002	All	88625–02010	2
	2003	CBU	88625–02050	2
	2003 – 2005	CBU	88625–02051	3
		NAP		
	2005	All	88625–02070	3
ECHO	2000 – 2003		88625–52020	2
	2003		88625–52100	2

**Inspection
Procedure
(Continued)**

MODEL	MODEL YEAR	COMMENTS	PART NUMBER	GRAPH
Highlander	2001 – 2005		88625-0C010	2
Land Cruiser	1997	40th Anniversary	88625-60060	2
	1998			
	1999 – 2003	Thermistor No. 1	88625-60130	2
	1998 – 2000	Thermistor No. 2	88625-60140	2
	2003 – 2005	Thermistor No. 1	88625-47011	2
	2000 – 2005	Thermistor No. 2		
Matrix	2003 – 2004		88625-02051	3
	2005		88625-02100	1
MR2	1990 – 1991		88625-24050	2
	1992 – 1995		88625-20350	2
	2000 – 2005		88625-17130	2
Paseo	1991 – 1995		88625-24050	2
	1995 – 1999		88625-12230	2
Previa	1990 – 1997		88625-60050	2
Prius	2000 – 2003		88625-20510	2
	2004 – 2005		88625-48050	1
RAV4	1996 – 2000		88625-20340	2
	2001 – 2005		88625-42080	2
Scion tC	2005		88625-21010	2
Scion xA	2004 – 2005		88625-52070	2
Scion xB	2004 – 2005		88625-52070	2
Sequoia	2001 – 2005		88625-0C010	2
Sienna	1998 – 2000		88625-16210	2
	2001 – 2003		88625-52020	2
	2004		88625-08010	3
		XLE (Rear)	88625-0C010	2
Solara	1999 – 2003		88625-33140	2
	2004 – 2005		88625-33170	3
Supra	1993 – 1998		88625-14120	2
T100	1993 – 1998		88625-34010	2
Tacoma	1995 – 2004		88625-35030	2
Tercel	1991 – 1994		88625-24050	2
	1995 – 1999		88625-12230	2
Tundra	2000 – 2002	All	88625-0C010	2
	2003 – 2004	All	88625-0C020	2
	2004 – 2005	Double Cab Only	88625-0C010	2
	2005	All		

**Inspection
Procedure**
(Continued)

A. Measure the sensor resistance according to the selected graph (specification).

NOTE:

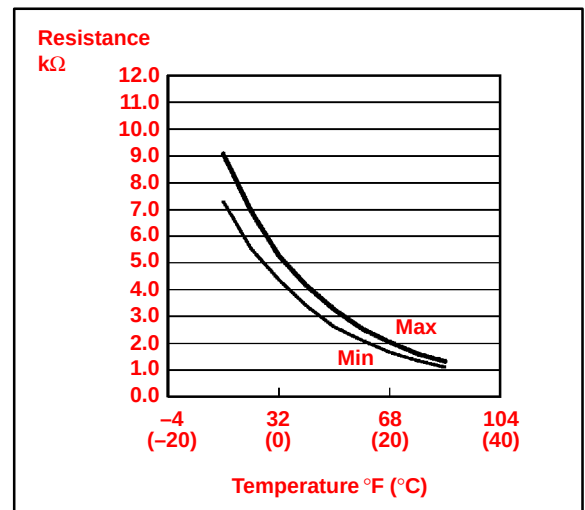
- Even slightly touching the sensor may change the resistance value. Be sure to hold the connector of the sensor.
- When measuring, the sensor temperature must be the same as the ambient temperature.

HINT:

As the temperature increases, the resistance decreases.

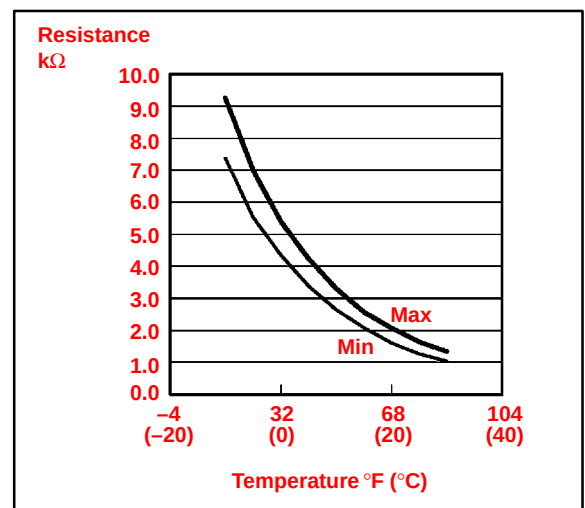
Graph 1:

TEMPERATURE °F (°C)	SPECIFICATION kΩ
14 (–10)	7.30 to 9.10
23 (–5)	5.65 to 6.95
32 (0)	4.40 to 5.35
41 (5)	3.40 to 4.15
50 (10)	2.70 to 3.25
59 (15)	2.14 to 2.58
68 (20)	1.71 to 2.05
77 (25)	1.38 to 1.64
86 (30)	1.11 to 1.32



Graph 2:

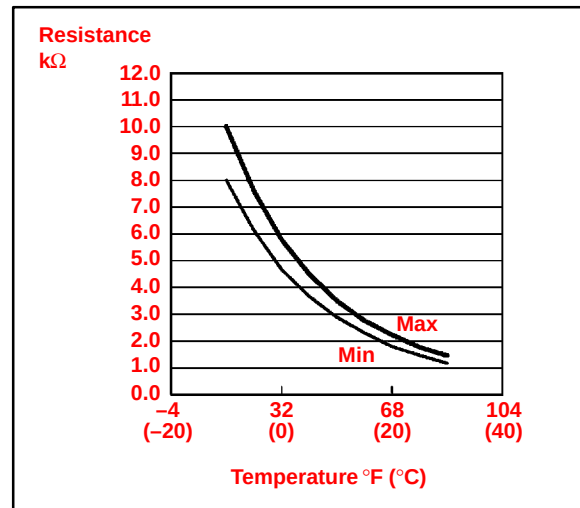
TEMPERATURE °F (°C)	SPECIFICATION kΩ
14 (–10)	7.40 to 9.20
23 (–5)	5.65 to 7.00
32 (0)	4.35 to 5.40
41 (5)	3.40 to 4.20
50 (10)	2.68 to 3.30
59 (15)	2.10 to 2.60
68 (20)	1.66 to 2.10
77 (25)	1.32 to 1.66
86 (30)	1.05 to 1.35



**Inspection
Procedure
(Continued)**

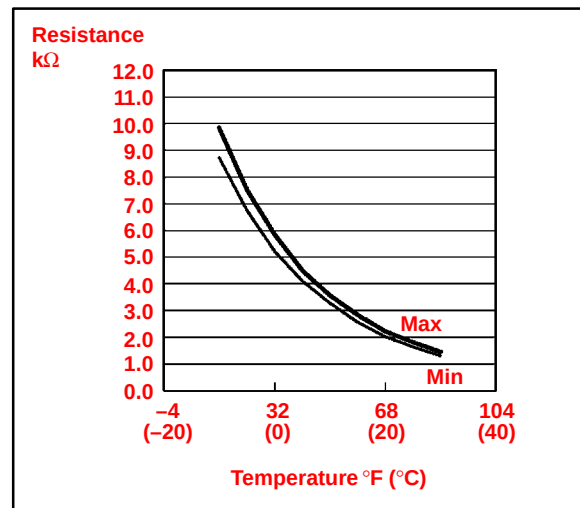
Graph 3:

TEMPERATURE °F (°C)	SPECIFICATION kΩ
14 (–10)	8.00 to 10.00
23 (–5)	6.15 to 7.65
32 (0)	4.75 to 5.85
41 (5)	3.70 to 4.55
50 (10)	2.91 to 3.55
59 (15)	2.32 to 2.80
68 (20)	1.85 to 2.22
77 (25)	1.48 to 1.77
86 (30)	1.20 to 1.43



Graph 4:

TEMPERATURE °F (°C)	SPECIFICATION kΩ
14 (–10)	8.80 to 9.85
23 (–5)	6.80 to 7.55
32 (0)	5.28 to 5.86
41 (5)	4.11 to 4.56
50 (10)	3.22 to 3.56
59 (15)	2.56 to 2.82
68 (20)	2.04 to 2.24
77 (25)	1.64 to 1.80
86 (30)	1.32 to 1.46



**Inspection
Procedure**
(Continued)

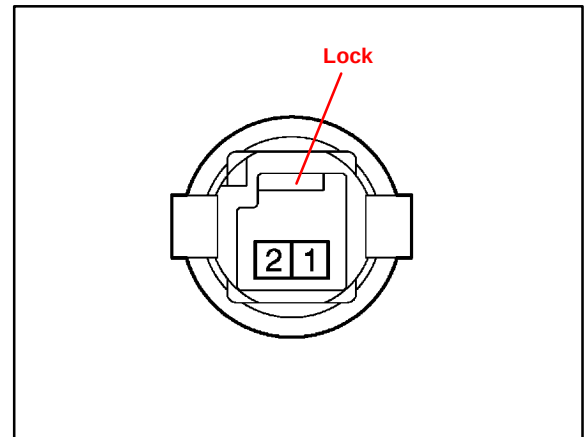
5. Inspect Solar Sensor.

Four types of solar sensors are used on Toyota vehicles depending on the vehicle specifications. The inspection procedure for each type of sensor differs from the others. Select the appropriate inspection procedure from the table below according to vehicle specifications and perform the inspection.

EQUIPPED WITH AUTOMATIC LIGHT CONTROL SYSTEM	A/C SYSTEM WITH RIGHT/LEFT INDEPENDENT TEMPERATURE CONTROL	INSPECTION PROCEDURE
No	No	A
No	Yes	B
Yes	Yes	C
Yes	No	D

Procedure A:

- a. Disconnect the solar sensor connector.
- b. Measure the resistance between terminals 1 and 2 of the solar sensor under the following conditions:
 - Cover the sensor with a cloth to avoid direct light.
 - Expose the sensor to light from a distance of 300 mm (11.81 in.) or less with an inspection light.



NOTE:

- Terminal 1 of the sensor is always on the right, when the lock is facing up.
- When using an analog tester, connect the positive (+) lead to terminal 2 and negative (–) lead to terminal 1 of the solar sensor.

HINT:

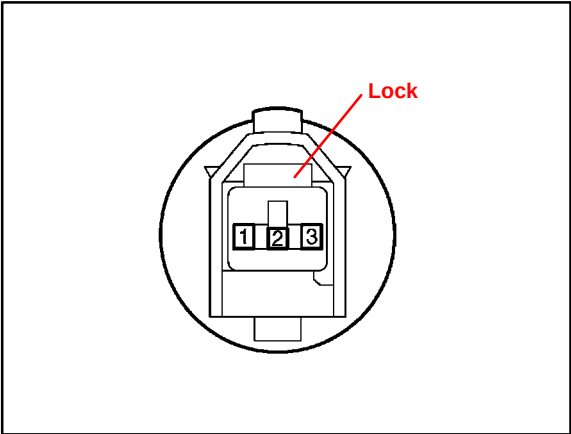
If the light is weak, the sensor may not react. Be sure to use an incandescent light for an inspection light.

Standard:

CONDITION	SPECIFICATION
When the sensor is covered with a cloth (to avoid direct light)	Infinite ohms
When the sensor is exposed to light	Less than infinite resistance

Inspection
Procedure
(Continued)

- Procedure B:
- a. Disconnect the solar sensor connector.
 - b. Measure the resistance between terminals 2 and 3 of the solar sensor under the following conditions:
 - Cover the sensor with a cloth to avoid direct light.
 - Expose the sensor to light from a distance of 300 mm (11.81 in.) or less with an inspection light.



NOTE:
When using an analog tester, connect the positive (+) lead to terminal 3 and negative (–) lead to terminal 2 of the solar sensor.

HINT:
If the light is weak, the sensor may not react. Be sure to use an incandescent light for an inspection light.

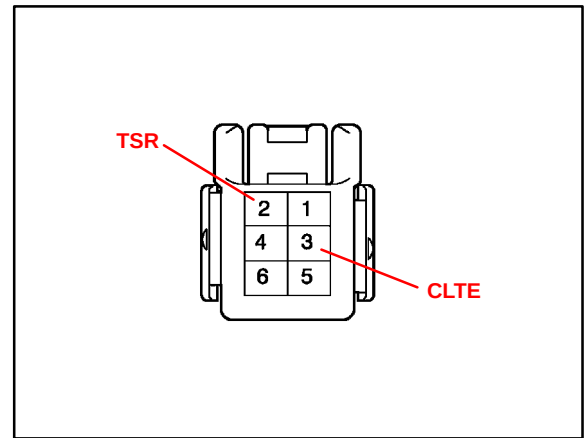
Standard:

CONDITION	SPECIFICATION
When the sensor is covered with a cloth (to avoid direct light)	Infinite ohms
When the sensor is exposed to light	Less than infinite resistance

**Inspection
Procedure**
(Continued)

Procedure C:

- a. Turn the ignition switch ON.
- b. Measure the voltage between terminals TSR (+) and CLTE (–) of the connector under the following conditions:
 - Cover the sensor with a cloth to avoid direct light.
 - Expose the sensor to light from a distance of 300 mm (11.81 in.) or less with an inspection light.



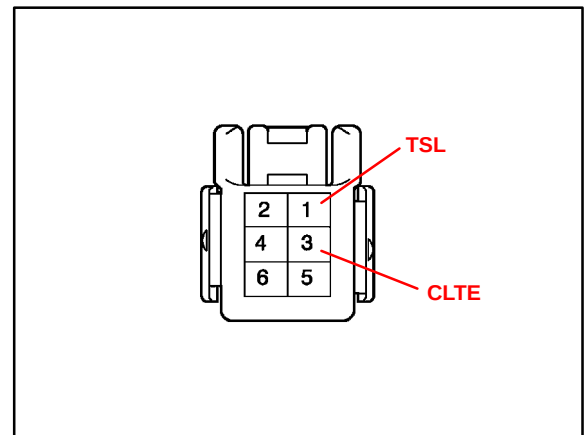
HINT:

- If the light is weak, the sensor may not react. Be sure to use an incandescent light for an inspection light.
- Do not disconnect the solar sensor connector.

Standard:

CONDITION	SPECIFICATION
When the sensor is covered with a cloth (to avoid direct light)	Below 0.8 V
When the sensor is exposed to light	4.3 +/- 0.3 V

- c. Measure the voltage between terminals TSL (+) and CLTE (–) of the connector under the following conditions:
 - Cover the sensor with a cloth to avoid direct light.
 - Expose the sensor to light from a distance of 300 mm (11.81 in.) or less with an inspection light.



HINT:

- If the light is weak, the sensor may not react. Be sure to use an incandescent light for an inspection light.
- Do not disconnect the solar sensor connector.

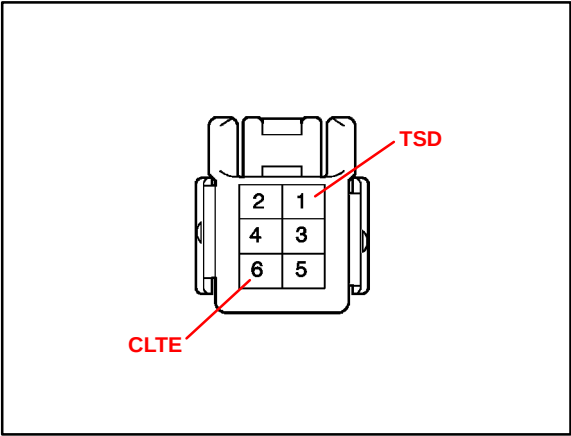
Standard:

CONDITION	SPECIFICATION
When the sensor is covered with a cloth (to avoid direct light)	Below 0.8 V
When the sensor is exposed to light	4.3 +/- 0.3 V

**Inspection
Procedure**
(Continued)

Procedure D:

- a. Turn the ignition switch ON.
- b. Using the tester, measure the voltage between terminals TSD (+) and CLTE (–) of the connector under the following conditions:
 - Cover the sensor with a cloth to avoid direct light.
 - Expose the sensor to light from a distance of 300 mm (11.81 in.) or less with an inspection light.



HINT:

- If the light is weak, the sensor may not react. Be sure to use an incandescent light for an inspection light.
- Do not disconnect the solar sensor connector.

Standard:

CONDITION	SPECIFICATION
When the sensor is covered with a cloth (to avoid direct light)	Below 0.8 V
When the sensor is exposed to light	4.3 +/- 0.3 V

**Inspection
Procedure**
(Continued)

6. Inspect Room Humidity Sensor.

Measure the humidity and output voltage of the humidity sensor when the sensor is installed on the vehicle and the temperature at the humidity sensor position (room temperature sensor position) is 77°F (25°C). If the output voltage is within the specifications according to the graph and table below, the sensor is normal.

HINT:

For the inspection procedure of the room temperature sensor, refer to “Room Temperature Sensor Inspection Procedure” in this bulletin.

- A. Turn the ignition switch to the ON position.
- B. Measure the voltage between terminal VO (3) and GND (2) of the room humidity sensor.
- C. Measure the humidity and voltage when the room temperature (humidity sensor position) is 77°F (25°C). According to the result, determine whether the sensor is normal or not.

HUMIDITY (% RH)	OUTPUT VOLTAGE AT 77°F (25°C)
10	0.70 to 1.08 V
20	0.72 to 1.57 V
30	1.13 to 1.95 V
40	1.61 to 2.24 V
50	1.99 to 2.46 V
60	2.26 to 2.66 V
70	2.48 to 2.85 V
80	2.68 to 3.04 V
90	2.87 to 3.05 V

